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UNDERSTANDING FOOD GROUPS IN THE FOOD-BASED DIETARY GUIDELINES: INSIGHTS FROM BRAZILIAN CONSUMERS

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Abstract:

In the Food-Based Dietary Guidelines (FBDG), the classification of foods is based on food groups and nutrient sources. Many studies have already investigated several aspects of consumers' understanding of the information contained in these documents. However, no studies have investigated consumer understanding of all foods in the groups listed in the FBDG. The purpose of this study was to assess Brazilian consumers' understanding of the classification of foods based on food groups in the FBDG concept. Therefore, a tool to assess consumer understanding of food groups, the Consumer Understanding of Food Groups (UFG), was developed and validated. This instrument included his 44 items that were approved by experts (agreement rate >80%). A total of 894 Brazilians from all regions participated in this study. This result suggests that 48.9% of participants thought it was easier to categorize foods into food groups. Food groups are classified according to the origin of the food (animal and plant). Although consumers can easily identify foods based on their origin, there is still an asymmetry in the inclusion of foods from the animal kingdom and species from the plant kingdom. This exploratory study reveals important information that may help improve FBDG. It is important to consider consumer understanding and guide consumer decision-making from a technical perspective.

Keywords : Food-based Dietary, Food classification, Plant kingdom

Introduction :

Dietary Guidelines guide consumer behavior based on national food, nutrition, and health guidelines. Aspects of healthy eating should be considered and scientific principles taken into account. Diet is known to be a factor that has an important influence on health. Improving dietary habits by reducing sodium intake and increasing intake of whole grains and fruit has the potential to significantly reduce morbidity and mortality from non-communicable diseases. In addition, limiting sugar and saturated fat and increasing intake of fiber and unsaturated fat also benefit public health. Diets that align with current evidence-based guidelines reduce the risk of all-cause and cause-specific mortality, so simultaneously guide food choices based on scientific evidence and consumer understanding of these choices [1].

In this scenario, classifying foods based on their physical, chemical, nutritional, biological properties and other food components will develop programs and policies in the fields of nutrition, health, agriculture and the food industry. A food classification is a distribution that lists various foods into groups, defined on the basis of common characteristics, and may include subgroups that are primarily identified by consumers. This grouping identifies a collection of foods that are not generally considered variations of the same food, but share common characteristics in terms of type, origin. According to the Food and Agriculture Organization (FAO) food classifications used in the Food-Based Dietary Guidelines (FBDG) are based on food groups and nutrient sources.

Food classification by food groups (including grains, fruits, vegetables, dairy products, and meat) classifies different foods according to their origin nutritional properties, and marketing characteristics. Therefore, the FBDG provides evidence-based, practical, and actionable recommendations aimed at guiding the population's dietary behavior through consumer education and targeted health policies and programs. The placement and classification of specific foods in Dietary Guidelines is based on the traditional dietary patterns of the country [2].

As a food and nutrition education tool for the public, the FBDG requires countries to describe foods using appropriate and understandable nomenclature, consider food classification from a scientific perspective, and use this information to It is necessary to "translate" it for people. Avoid contradictory and ambiguous information [2]. Researchers have already investigated consumer understanding of the graphic information and information contained in the FBDG, as well as the intake of specific food groups included in these guidelines, including: There is. B. Comparison of information between milk and vegetable groups and his FBDG of several countries. However, no study has assessed consumers' understanding of all the foods included in their group and listed in his FBDG available on the FAO website. Therefore, this study aimed to assess Brazilian consumers' understanding of the concepts of food classification by food groups and food-based dietary guidelines [3].

Materials and methods:

Research Design:

This qualitative and quantitative cross-sectional study was conducted among Brazilian consumers. The first step was the development and validation of the Consumer Understanding of Food Groups (UFG) tool. Instrument validation was performed carried out. [33]. UFG underwent content and semantic analysis. The Ethics Committee of the University of Brasilia UFG was sent electronically to consumers in all geographic regions of Brazil [4].

Structure of the Understanding Food Groups (UFG) tool:

The proposed elements of this tool follow the nomenclature and food classifications described in the 89 food-based dietary guidelines available on the FAO website. was created. The preliminary version contained 66 items (components). Fifty-nine experts (university professors, food technologists, food scientists, and nutritionists) were invited to assess the clarity and relevance of the items. They were also asked to submit suggestions for corrections or additions to the article if necessary. Twenty-three experts agreed to participate in this part of the study and rated the items using the following criteria: (i) Item clarity was assessed using a 5-point scale: 1 (I didn't understand it at all), 2 (I didn't understand it at all). I understood a little), 3 (I understood almost everything, although I had reservations), 4 (I understood almost everything), and 5 (I completely understood). (ii) Item relevance was rated on a 5-point scale: 1 (inadequate), 2 (not sufficient), 3 (not sufficient), 4 (sufficient), and 5 (very sufficient) [5]. The data were analyzed considering each item with a score of and the suggestions for reformulating the item presented. Two university researchers from Food Science, Technology and Nutrition evaluated the paper and revised it according to expert suggestions. The criterion for retaining items in the instrument was to achieve at least 80% inter-expert agreement for each item. The items that did not reach 80% were revised and reevaluated. At the suggestion of experts, this element was excluded (Figure 1).

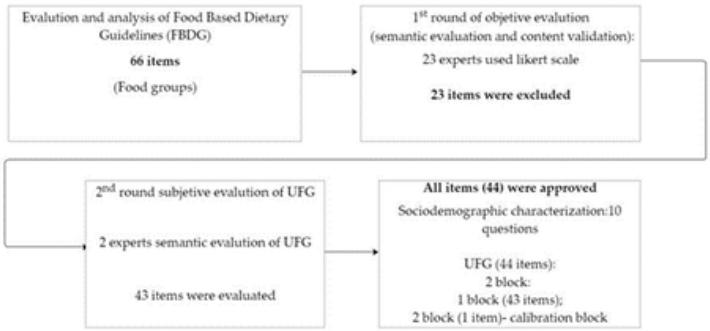

Figure 1: Stages of the construction, content validation, and semantic evaluation of understanding of Food Group.

The final version of the UFG included 43 foods and food groups. Foods are described according to the information available in the FBDG, and some only include photos of foods or items representing specific groups. This is because the FBDG provides practical, evidence-based, and actionable recommendations for managing dietary behaviors [6]. Respondents were asked to respond by rating items as "true," "false," or "don't know" for foods and food groups (table 1).

Items	True		False		I Do not Know	
	n	%	n	%	n	%
I-15. Yogurt, cheeses, and cottage cheese are dairy products ("produtos lácteos").	882	98.7	10	1.1	2	0.2
I-18. Sweet potatoes, carrots, cassava, and turnips are roots.	832	93.1	48	5.4	14	1.6
I-19. Curd is a dairy product.	879	98.3	6	0.7	9	1
I-20. Coconut and avocado are fruits.	823	92.1	45	5	26	2.9
I-31. Rabbit, wild boar, snake, and alligator meat are game meats.	808	90.4	59	6.6	27	3
I-32. Yogurt, cheese, and cottage cheese are derived from milk.	887	99.2	6	0.7	1	0.1
I-37. Lettuce, arugula, and kale are leafy foods.	865	96.8	12	1.3	17	1.9
I-40. Yogurt, cheeses, and cottage cheese are dairy products ("laticínios")	843	94.3	37	4.1	14	1.6
I-42. Oils, olive oils, butter, and margarine are fats and oils.	842	94.2	32	3.6	20	2.2

Table 1: The numbers and frequency of responses indicated by respondents as "True" with a frequency greater than 90%.

To analyze the results, we considered the percentage of true or false and don't know responses. We compared these rates with sociodemographic data and botanical, agronomic, and nutritional characteristics of each item. In addition, items that examine why consumers feel that processing level food group nutrient source and ingredient list are easier when classifying foods Also added. If this is the consumer response to the food classification previously validated and applied to the Brazilian population. Additionally, sociodemographic data such as nationality, administrative region, country of residence, age, gender, education level, and monthly income are included (table 2) [7].

Items Included in the Vegetal Groups	Characteristics	Category	% True	*p*-Value
Green banana is a type of cereal (I-2)	Educational level	Up to high school	9.3	$p = 0.001$ $p < 0.05$
		College degree	6	
		Postgraduation	3.5	
	Income—minimum wage (BRL 1100) USD 1 = BRL 5.16	Up to four	6.8	$p = 0.006$ $p < 0.05$
		From five to nine	4	
		Above 10	2.8	
		Not informed	4.7	

Table 2: Consumer understanding of UFG items.

The UFG was distributed nationwide using a random sample using the snowball method. The instrument validation process requires 20 respondents per item (20:1). Therefore, to validate the UFG, the minimum sample size was estimated to be 860 (20:43) participants [1]. The instrument was applied to a sample of adults from all regions of Brazil using the Google Forms™ platform (Menlo Park, CA, USA). Participants were recruited through social media advertisements (Menlo Park, CA, USA Facebook™, Menlo Park, CA, USA Instagram™, and Menlo Park, CA, USA WhatsApp™ Menlo Park). Data collection took place from February 2022 to December 2022[8].

Statistical analysis:

Categorical variables (sociodemographic characteristics) were described as frequencies (n) and percentages (%), and quantitative variables were described as mean and standard deviation or standard error. To examine differences in results, independent Student's t-test, His ANOVA with Tukey post hoc test, and ANOVA with Tukey post hoc test were used. Categorical variables were compared using the chi-square test. The statistical significance level was set at 5% ($p < 0.05$). The statistical software IBM SPSS Statistics for Windows version 22 (IBM Corp, Armonk, NY, USA) was used for the analysis [9].

Results:

Understanding Food Groups (UFG): Of the 902 people who accessed the applied UFG, 99.1% (n = 894) agreed to participate in this study and signed a free informed consent. did. Most of the participants were from the Southeast region of Brazil (n = 348; 38.92%), followed by the Northeast (n = 208; 23.2%), the North (n = 120; 13.4%), and the Midwest (n = 118; 13.2%). and the South (n = 100; 11.2%) (Appendix B). Most of the respondents were women (n = 559; 62.5%), and more than half were 4,444 (n = 528; 59%; n = 528) over the age of 40. Most respondents had an education level equivalent to a graduate degree (n = 683, 76.4%), and most respondents' personal income was above 10 minimum wage (n = 353, 39.5%) [10].

Added to the group of vegetables and fruits according to FBDG. For Block 1, data on the frequency of responses of True or False and I don't know are provided in. Of the 43 items, only 9 items (21%) were reported as True, False, and True unknown, with a frequency of 90% or more. Of these, 98.7% (n = 882), 98.3% (n = 879), 99.2% (n = 887), and 94.3% (n = 843) of respondents scored points I-15, I-19, I-32 was evaluated. I-40 (quark, yogurt, cheese, cottage cheese) belongs to the group of dairy products [4]. Items I-18 (sweet potatoes, carrots, cassava, and beets are roots) (n = 832; 93.1%); I-20 (coconuts and avocados are fruits) (n = 823; 92.1%); I-31 (rabbits, Boar, snake, and crocodile meat are game meat) (n = 808; 90.4%), I-37 (lettuce, arugula, and kale are leafy foods) (n = 865; 96.8%), and I -42 (Oil, olive oil, butter, and margarine are fats and oils) (n = 842; 94.2%) was also reported as True with a frequency of more than 90% of responses. More than 90% of respondents answered two items: I-22 (Nuts, chestnuts, and seeds are "vegetables") (n = 825, 92.3%) and I-24 (Seafood, poultry, vegetables, eggs) was answered as "incorrect", dairy products are meat) (n = 807; 90.3%) [11].

Sociodemographic data, frequency of answers reported as correct and importance levels for items related to meat groups. Of his 12 items in the meat group, 9 items were significantly different ($p < 0.05$) with respect to the variables of gender, age group, geographic region, education level, and income [7].

Regarding gender, I-1 and I-31 showed significant differences between answers reported as "correct." Female respondents said that "meat is the edible parts of animals (n = 257, 46%) and rabbit, boar, snake, and crocodile." Meat was game (n = 520, 93%). Significant differences were also observed in age groups regarding I-43 (meat included in cattle, pigs, sheep, goats, buffalo, poultry, fish, amphibians, mammals, and reptiles): >70% (n = 373) respondents aged 40 and older rated this statement as true.

Items I-9 and I-34 related to fishes and fish groups also show large differences in age groups, geographic regions, education levels, and incomes. Item I-9 had the highest percentage of responses marked “incorrect” (n = 720, 80.5%). Respondents from the Northeast region (n = 20, 9.6%) with higher education (n = 21, 12.5%) and earning up to four times the minimum wage (n = 24, 10.8%) said, “Algae These were the people who said, "There is." Set "a Seafood (I-9)" to "True". Regarding item, I-34 (freshwater fish and saltwater fish), the percentage of respondents aged 40 and over who answered that they had no income (n = 470, 89%) (n = 39, 90.7%) answered “agree” was high [12].

For items I-11 (meat is beef) and I-24 (seafood, chicken, vegetables, eggs, and dairy products are meat), significant differences were found in the education level and income variables. Regarding item I-11, 770 (86.1%) of the respondents' girlfriends answered that item as "false." Respondents with higher education (n = 32, 19%) and earning up to the $4,444 minimum wage (n = 41, 18.5%) reported the highest percentage of “true” responses for this item. It was the people. Regarding item, I-24, a high percentage of respondents answered “wrong” (n = 807, 90.3%). Respondents whose educational background was less than high school (n = 5, 11.6%) and whose income was not specified (n = 4, 9.3%) had a higher percentage of responses marked as true. Papers. Items I-5 and I-21 related to the meat product group showed significant differences in terms of gender, age group, and geographic region. Male respondents (n = 306, 91.3%) said "correct" about point I-5 (hot dogs, sausages, ham and mortadella, among others, are meat products), and respondents aged 40 and older (n = 478, 90) were similar, 5). %) and residents of the southern region of Brazil (n = 93, 93%). Among male respondents, 170 (50.7%) reported item I-21 (meat such as hot dogs, sausages, ham, and mortadella) as true, as did residents of the southern region (n = 51; %) [13].

Frequency of responses reported as True (Block 1), importance, and sociodemographic data related to items related to the dairy group. Of the 8 items described in this group, ``5 of them had significant differences ($p < 0.05$) in the following variables: gender, age group, geographic region, education level, and income.'' I-8 (Kefir There were significant differences in the frequency of responses reported as True for gender, age group, and geographic region variables between I-23 (tofu is cheese) and I-23 (tofu is cheese). Female respondents (n = 237, 42.4%, n = 248, 44.4%) and people over 40 years of age (n = 201, 38.1%, n = 239, 45.3%) scored I-8 and I-23. Respondents living in the southern region (n = 47, 47%) reported a higher percentage of items I-8 to I-23 answered “true”.

The highest proportion of items reported as True was found in the Northern region (n = 60, 50%). Significant differences existed in I-19, I-13, I-40 and all variables. Female respondents (n = 553, 98.9%), 40 years of age or older (n = 525, 99.4%), high school education level (n = 43, 100%), women who did not declare their income (n = 43), 100%), reported a higher percentage of elements to be "true" for I-19 (the curd is derived from milk). Female respondents (n = 216, 38.6%), up to 39 years of age (n = 144, 39.3%), resident in the northern region (n = 54, 45%), with higher education (n = 73, 43.5) %), those with up to 4 minimum wage incomes (n = 98, 44.1%) had a higher percentage of points answering “correct” on I-13 (soy milk and rice milk are milk). Item I-40 (yoghurt, cheese, and cottage cheese are dairy products) was reported as “true” among respondents who were 40 years of age or older (n = 507, 96%) and had graduate-level education. The frequency of the items has increased. (n = 652.95.5%) [8].

Sociodemographic data for the items related to the grains, pulses and vegetable groups, along with the frequency of responses labeled true and the significance level. There were significant differences ($p < 0.05$) in the following variables: gender, age group, geographical region, education level, and income for 10 of the 12 items listed in the cereals, legumes, and vegetables group. Male respondents had higher item frequencies for I-12 (oats, rice, wheat, corn, grain seeds; n = 303; 90.4%) and I-35 (gnocchi, pasta, lasagna, cannelloni). True he reported. Obtained from grains. n = 209, 62.4) [5]. For women (n = 329, 58.9), the percentage of points reported as “true” for I-39 (Corn, wheat grain and other grains as grains or cobs belong to the group of cereals and legumes) A higher frequency was observed n = 329, 58.9 %.

In I-2 (green bananas are a type of grain), there were significant differences in education level and income variables. Respondents with a high school education level (n = 4, 9.3%) and an income up to four times her minimum wage (n = 15, 6.8%) were more likely to answer "correct" Ta. For I-33 (soybeans and chickpeas are “legumes”), there were significant differences in gender and age group variables. Male respondents (n = 104, 31%) and respondents whose girlfriends were 40 years of age or older (n = 162, 30.7%) were more likely to answer “true.” For I-14 (soy steak is a type of meat), significant differences were observed across geographic region, education level, and income variables. Respondents (n = 47, 21.2%) who reside in the Midwest region (n = 30, 25.4%), have a higher education (n = 41, 24.4%), and earn up to four minimum wages (n = 47, 21.2%) showed higher values of frequency. There were more answers than correct [14,15].

Respondents answered "true" to I-41 (Beans of all colors, fava beans, lentils, and peanuts are legume seeds) more often than articles describing such foods (n = 666; 74.5%). Grains and vegetables or legumes. Regarding the legume group, there were significant differences in all variables I-16, I-27, I-36, and I-41. (n = 175, 52.2%, n = 222, 66.3%) and respondents aged 40 or older (n = 89, 54.7%, n = 340, 64.4%) answered yes or no about I. "Yes," was the most frequently answered answer. -16 (beans, soybeans, lentils, and chickpeas are grains) and I-27 (beans, soybeans, lentils, and chickpeas are vegetables). There were also significant differences in education level and income variables in I-27. Respondents with a graduate degree (n = 421, 61.6%) and those who did not report income (n = 28, 65.1%) were more likely to answer true.

I-36 (beans, soybeans, lentils, chickpeas, legumes) had significant differences in geographic region, education level, and income variables. In I-41, there were significant differences in age group and geographic region variables (beans of all colors, broad beans, lentils, and peanuts are seeds of the legume family). Residents of the southern region answered “true” more frequently for items I-36 and I-41, at 73% and 85%, respectivelys (n = 261, 73.9%) with a graduate degree (n = 485, 71%) and earning more than 10 times minimum wage answered “true” about the I-36 were more likely to answer. For I-41, respondents aged 40 and older (n = 410, 77.7%) were more likely to answer true [10].

Sociodemographic data, frequency of responses reported as “true” (Appendix C), and importance levels for items related to fruits, oils, fats, and vegetable groups. Among his 11 items in the fruit, oil, fat, and vegetable groups, all variables in 8 items had significant differences ($p < 0.05$). Only one of the two fruit-related items (I-7 and I-20) was significantly different with respect to the geographic region variable. Respondents in the Southeast region (n = 305, 87.6%) reported answering “true” more frequently for I-7 (tomatoes, melons, watermelons, green peppers, and bell peppers are fruits).

Female respondents (n = 453, 81%) were more likely to answer “true” for I-6 (garlic and onions are onions) [3]. In contrast, male respondents (n = 199, 59.4%) were more likely to answer “true” for I-10 (vegetables are leaves, flowers, fruits, stems, seeds, tubers, and roots). There were also significant differences in geographic region variables in I-10. Residents of the Midwest region reported answering “true” more frequently (n = 82, 69.5%). For articles I-4 (Cauliflower, broccoli, and artichokes are flowers), I-6 (Garlic and onions are onions), I-22 (Nuts, chestnuts, and seeds are vegetables), I-25 (" vegetables").

There were significant differences in age group variables for I-37 (edible green leafy parts of plants) and I-37 (lettuce, arugula, and kale are leafy foods). For all of these questions, respondents aged 40 and older reported a higher frequency of “true” answers 67%, 85.2%, 5.7%, 66.7%, and 97.2%, respectively.

For items I-6 and I-25, we also found significant differences in geographic region and income variables. Residents of the southern region answered “true” more often, at 84% and 64%, respectively. For I-6, we observed that respondents who did not report income (n = 37, 86%) were more likely to answer “true.” For I-25, respondents who earned more than 10 times her minimum wage (n = 226, 64%) were more likely to answer “true”.

There were significant differences in the education level variables for items I-4, I-18 (sweet potatoes, carrots, cassava, and beets are roots), I-22, and I-25. Respondents with graduate education voted I-18 (n = 641, 93.9%), I-22 (n = 31, 4.5%), and I-25 (n = 421, 61.6%). Respondents who studied through high school were more likely to answer “true” about I-4 (n = 34, 79.1%) [2,9].

For block 2 data, the objective was to determine whether (1) "level of processing", (2) food group, (3) nutrient source, or (4) ingredient list, there were significant differences between respondents by age. Only layer variables. In general, the results show that most respondents think it is "easier" to categorize foods by food groups (n = 437, 48.9%), and that respondents aged 40 and above There was a significant difference between (n = 270, 51.1%) [13,14].

Discussion:

This study, which recruited respondents from all Brazilian states, is the first to assess consumers' understanding of the classification of foods by food groups listed in the Food-Based Dietary Guidelines on the FAO website. research. Our UFG validation was performed. Regarding the number of experts required to assess the suitability of each element, there is no consensus on this number in the literature.

The number of respondents was his 894, which is consistent with the recommendation goes on to validate the instrument with a minimum number of 20 respondents per point. In this study, similar to other studies, most respondents were female (62.5%). Women participate in online surveys more often than men, and women are known to have healthier eating habits than men.

For the 43 items (100%) that make up the UFG, the frequency of "true" responses varied between 3.8 and 99.2%. The frequency of "false" responses varied between 0.7 and 92.3%, and the frequency of don't know responses varied between 0.1 and 36.9%. In addition to animal-based foods such as dairy products and some wild species, plant-based items (coconuts, avocados, lettuce, arugula, kale) and fats and oils (of animal and plant origin) were also on display. It was rated "true" by respondents.

Grouping foods according to their origin, such as animal or plant, was one of the first strategies for classifying foods, and is therefore a type of classification that is familiar to most consumers. Regarding origin, some his FBDG further divides foods of animal origin into two groups: One group refers to meat and eggs, and the other refers to milk and dairy products.

This is probably because they are a source of iron and calcium, respectively. To ensure that respondents do not have any doubts about the meaning of the terms 'food category' and 'food group', this tool consists of the following items I-1 -Meat is the edible part of an animal responses were grouped by researcher. We were divided into our respective groups. This is because foods can be classified according to various criteria origin, nature, processing, nutrient source, food group, etc.

We conducted a study according to a previously published and in the present study, most respondents (48.9%) believe that it is easier to classify foods according to food groups. However, we found that among the investigated FBDGs, those using food classification based on food groups do not have a logical structure. This is because items such as green bananas, coconuts, avocados, dairy products, and legumes are reported in different food groups. When nutritional and botanical standards are different. Significant differences have been observed in the grouping of species in the plant kingdom.

Vegetables are classified in terms of edible parts: roots, tubers, rhizomes, leaves, fruits, soft stems and shoots, inflorescences flowers, bulbs, immature seeds, and cultivated mushrooms. For this reason, lettuce, kale, chicory, cabbage, watercress, and spinach are leaves. Yams, challah, and potatoes are tubers, while taro is a rhizome. Onions and garlic are onions. Cauliflower, broccoli and artichoke are inflorescences flowers. Zucchini, okra, eggplant, green corn, tomatoes, pumpkin, melon, watermelon, and strawberries are fruits. Green beans, kidney beans, peas, and snap peas are immature pods or seeds.

Carrots, cassava, turnips, radishes, turnips, sweet potatoes, yacon, and parsnips are roots; celery, asparagus, and cellulose are soft stems. From a nutritional point of view, edible parts such as tubers, roots, and rhizomes contain higher levels of energy stores (carbohydrates). Kidney beans or kidney beans, on the other hand, can contain up to 40% protein in their chemical composition, depending on the variety. Fruits, leaves, and onions are sources of bioactive compounds such as vitamins, minerals, and flavonoids. For grains and legumes, it is also important to consider their chemical properties to estimate their nutritional value. These factories are grain seed suppliers. Grains/seeds such as rice, wheat, corn and their derivatives are sources of complex carbohydrates. Legumes such as ripe kidney beans, lentils, fava beans, chickpeas, peanuts, and soybeans are sources of protein and carbohydrates.

Oilseeds are plants that contain high oil content in both the seeds (soybean, rapeseed, sunflower, castor, jatropha, crambe, chia) and fruits (palm, babassu, coconut). Considering the protein content, soybeans can be classified as legumes. However, if the parameter is industrial processing, soybean is classified as an oilseed (fat source). The results showed that Brazilian respondents reported item 22 (Nuts, chestnuts, and seeds are "legumes" or vegetables) as "incorrect" (n = 825, 92.3%).

Because more than 60% of the respondents were beans, soybeans, lentils and chickpeas. It is a legume or a seed of the Fabaceae family. Beans and other legumes are foods that, according to the FBDG, can be classified in the fruit and vegetable group, in the cereal group, and even in the meat group, or together with oilseeds. As already emphasized, these foods differ in chemical composition and nutritional properties.

A study surveyed 314 Australian respondents' understanding of the terminology used in the heading 'legumes and whole grains' in their country's dietary guidelines. The results showed that the majority of respondents (n = 123, 45%) responded that legumes should be included as part of their group or protein source food group. According to Codex Alimentarius, legumes, nuts and seeds are classified into specific groups.

According to the results, more than 90% of Brazilian respondents understood item 24 (seafood, poultry, vegetables, eggs and dairy products are meat) as "false" (n = 807, 90.3%). This sample shows: Brazilians eat legumes, not meat (I-24). More than 80% of Brazilian respondents reported item 14 (soy steak is a type of meat) as incorrect (n = 751, 84%).

Therefore, taking into account the results and data of previous studies to understand the group to which legumes should belong, and the nutritional and botanical properties of these foods, the FBDG has placed the following on items belonging to the legume group: You need to create specific groups for them. Understanding refers to an agreement reached about an event or issue.

Despite the close relationship, knowledge does not mean understanding. Understanding is the act of interpreting, evaluating, or recognizing what is discussed, debated, and specifically disclosed. These are important for estimating the value of an attribute based on its cause. Consumers can make poor decisions if they have no idea what's at stake. In this way, various food classification proposals have emerged, as well as food classification systems aimed at linking food groups and dietary habits with socio-demographic data and diseases.

Although some studies have assessed the consumption of specific food groups based on sociodemographic data and diseases, few studies have been published in the literature regarding knowledge understanding of food groups and sociodemographic data.

Reported that female gender, family education, and high income positively influenced fruit and vegetable consumption among Canadian adolescents. In this study, male respondents in the Midwest region had a higher response rate (59.4%) for item I-10 (vegetables are leaves, flowers, fruits, stems, seeds, tubers, and roots). Investigated the food consumption of Brazilian adults in urban and rural areas and concluded that fruit and vegetable consumption is lower in the Northeast macroregion compared to consumption in other regions of Brazil.

Fruits are essential to a healthy diet because they contain beneficial non-nutritive substances such as vitamins, minerals, fiber, and bioactive compounds. The World Health Organization (WHO) recommends consuming at least 400g (about 5 servings) of fruits and vegetables each day. Low fruit intake is one of the main risk factors for increased mortality and also increases the risk of chronic diseases and poor health. Brazilian respondents from the Southeast region reported a high percentage of correct answers (87.6%) for I-7 (tomatoes, melons, watermelons, bell peppers, green peppers are fruits). The Southeast region accounts for 40.87% of Brazil's fruit and vegetable production [68]. From a botanical point of view, tomatoes are fruits. Consumers commonly recognize this food product as “Verdura”.

Baek and Chitekwe studied the food consumption of children and adolescents in Nepal and observed that there were differences in the consumption of various foods, including dairy products, depending on geographic location and income. This study found significant differences in items I-8, I-13, and I-23 for geographic region criteria. According to Zoccal the northern region is the region with the lowest milk consumption, and respondents from the northern region answered “true” at a higher rate for items I-13 (soy milk and rice milk) and I-13. This may explain why you reported it -23 (Tofu is cheese). Studied the dairy consumption habits of Brazilian and Portuguese adults. The results showed that approximately half (46.7%) of Brazilian respondents had never consumed semi-skimmed milk powder, and only 1% (46.7%) of Brazilian respondents had ever consumed semi-skimmed milk powder; We found that the number has increased. 3%).

In both countries, the number of participants consuming imported cheese is particularly low (only 4.0% of participants consume these products at least once a week), suggesting that domestic products may be preferred. They also found that people who consumed cheese either rarely (once a week) or only sometimes (two to three times a week). Yogurt consumption also tends to be low in the most commonly consumed yogurt types in Brazil are creamy pulp yogurt (14.4% regularly consumed), liquid yogurt (13.7% regularly consumed), and Greek yogurt (10.2% regularly consumed).

Found that despite some differences in dairy consumption behavior between the two countries (Brazil and Portugal) dairy product consumption levels were high for all items assessed (milk, cheese, yogurt, butter). Concluded that it was very low. Examined the differences in fish consumption between macro-regions in Brazil and found that fish consumption was higher in the northern and northeastern regions. In this study, respondents in the Northeast region reported a higher percentage of "correct" answers for item I-9 (Seaweed is seafood) (9.6%, compared to 6.4% overall). This means that even though the Tohoku region is the largest marine area in Japan, over 80% of respondents do not understand that seaweed is not seafood or that seaweed can be consumed as food. This is thought to be because seaweed is easier to obtain.

Approximately 13% of respondents, regardless of region, indicated that they did not know how to answer item I-9. As a result, respondents from the southern region found that his I-36 (beans, soybeans, lentils, chickpeas are legumes) (73%) and I-41 (beans of any color, fava beans, lentils, peanuts are legumes) It turns out that the seeds of the legume family were reported.

(85%) answered "truth" also found that adults in the Southeast and Midwest regions consumed more beans, justifying the association between understanding and consumption.

It was also observed that foods such as insects, kefir, and green bananas had low consumption and were largely unknown to the Brazilian respondents of this study. His FBDG for Vietnam, South Korea, Cambodia, and Kenya included insect consumption. FAO defines all terrestrial insects such as spiders, mites, ticks, beetles, flies, beetles, ants, and maggots such as earthworms, including processed products such as dried insects and industrial products such as insect powder and maggots. Defined. It is included in the trophic subgroup "insects and maggots" [50].

More than 60% (n = 544) of the Brazilian respondents answered "false" to item I-17 (Insects such as grasshoppers, ants, maggots, cicadas, and dragonflies are meat), indicating that this Brazilian Human samples show that insects are not considered meat. Approximately 18% (n=159) answered "I don't know" regarding the same item.

Kefir is a low-alcohol fermented beverage with a sour taste and fizz made by fermenting kefir grains with milk or water. In his FBDG for Colombia, Estonia and Hungary, kefir was counted as a dairy product [17,34]. Approximately 38% (n = 338) of Brazilian respondents answered "true" for item I-8 (kefir is a dairy product), 4,444 answered "false" and 25% (n = 226) of them answered "false"; 37% (n = 330)). ``I don't know," he said. Women had the highest percentage of "true" responses, as did respondents in the southern region. Female respondents are most concerned with aspects related to health and the nutritional value of food. The FBDG of 17 Latin American countries has added green bananas to the group of cereals and legumes. Approximately 87% (n = 774) of respondents answered her I-2 (green bananas are a type of grain) as "incorrect." Respondents with an education level up to high school and an income up to four times her minimum wage reported the highest percentage of "correct" answers on I-2.

These data indicate that lower-educated and lower-income respondents in this study's sample understand that green bananas are a type of grain, whereas respondents from other sociodemographic groups do not. A green banana may indicate that it belongs to a group of fruits. Green bananas are considered to be a good source of fiber, vitamins, and bioactive compounds such as phenolic compounds and resistant starch (RS), which may contribute to health benefits.

The results showed that women had a higher proportion of items related to spices (I-6: Garlic and onions are bulbs, 81%) and prepared products containing plant extracts (I-13: Soy milk and rice). I did. true" Milk is milk, 38.6%. I-23: Tofu is cheese, 44.4%. Milk (I-8: Kefir is a dairy product, 42.4%; I-19: Cottage cheese is a dairy product, 98.9%) and meat (I-31: Rabbit, boar, snake, crocodile meat is game meat). Cooked meat, 93%, I-1: Meat is the edible part of an animal, 46%). Food selection research shows that women are generally responsible for preparing meals at home and selecting food for their families.

Conclusion:

This study provided a valuable international overview of the 89 FBDGs available on the FAO website. This data suggests that consumers find it easier to categorize foods by food group. Although consumers have traditionally been able to easily identify foods based on their origin (animal or plant), asymmetries still exist between foods within species groups of the animal kingdom and species groups of the plant kingdom.

In studies investigating the relationship between food consumption and health, it is important to emphasize that misclassification of foods can lead to erroneous interpretations. We conclude that this exploratory study reveals important information that may help improve the dietary guidelines outlined in the FBDG.

It is important to consider consumer understanding and guide consumer decision-making from a technical perspective. why? This is because food is an important source of nutrients that sustain our vital activities, and the balance between the quantity and quality of nutrients ensures optimal health.

Therefore, it is important to disseminate accurate information, always taking into account cultural characteristics, socio-economic characteristics, educational levels, etc. Therefore, the foods listed in the FBDG should be redistributed to their respective groups so that differences that are already common in research on food consumption and its contribution to health are minimized. We believe that new research is needed in other countries that assesses consumer understanding of food groups, as well as specific groups and food consumption, with the aim of improving consumer compliance with dietary guidelines. I have confirmed.

Conflict of interest :

The authors declare no conflicts of interest.

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